Joseph Yanney-Ewusie Newton

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LinkedIn, GitHub, Portfolio

Environmental sustainability graduate (MSc, TH Bingen) with a BSc in Agricultural Engineering, working on climate and agri-food data. Experience in GHG accounting, carbon MRV and geospatial analysis, built through independent projects and data quality work at BASF Digital Farming. Draws on hands-on agricultural work in Ghana and Israel and data roles in Germany to turn messy field, supply-chain and emissions data into practical answers for sustainability decisions.

Core Competencies

- **Carbon & Compliance**: GHG Protocol Scope 1 & 2, IPCC emission factors, carbon MRV, REDD+ / Verra (VM0048), VSME/ESRS mapping, Monte Carlo uncertainty analysis, Net Zero frameworks.
- **Geospatial & Remote Sensing**: Google Earth Engine, Sentinel-2, Landsat, Hansen GFC, land-cover classification (Random Forest), change detection, QGIS, raster and vector analysis.
- **Programming & Data**: Python (pandas, numpy, rasterio), SQL, PostgreSQL, ETL pipelines, time-series analysis, data quality validation and documentation.
- **Reporting & Visualization**: Power BI, Tableau, Streamlit, Excel, Matplotlib.
- **Tools & Collaboration**: Git/GitHub, AI-assisted development with Claude Code, Jira, Confluence, Microsoft 365.

Professional Experience

HTWest GmbH — **Bonn, Germany**
Werkstudent, Production Assistant — *Apr 2026-Present*

- Support production, packaging and shipping of precision-cut specialty papers and films, including Parafilm, filter media and wrappers, for B2B customers.
- Check orders, labels and finished products.

BASF Digital Farming GmbH (xarvio) — **Cologne, Germany**
Werkstudent, Data Quality (Drone Data & Sustainable Weed Management) — *Oct 2024-Feb 2026*

- Validated and documented drone datasets for weed-detection models, supporting reliable model outputs and operational decisions.
- Coordinated with teams in Germany, Argentina and Brazil to resolve data quality issues and align on process standards.
- Wrote stakeholder summaries and status updates, tracked recurring quality issues and supported corrective actions.

Technische Hochschule Bingen — **Bingen am Rhein, Germany**
University Research Assistant — *Sep 2023-Dec 2023*

- Collected, validated and structured field sensor data for climate adaptation research on crop moisture stress and temperature impacts.

Freezelink Cold Chain Logistics — **Accra, Ghana**
Project Analyst — *Jan 2022-Aug 2022*

- Compiled benchmark reports on cold chain performance across client sites, used to guide implementation and operational improvement decisions.
- Supported the planning and rollout of modular cooling solutions, coordinating information between project teams.
- Maintained structured project documentation and produced efficiency reports for client reviews.

Tzafarti Ibudim (Citrus) — **Moshav Ahituv, Israel**
Trainee — *Dec 2020-Oct 2021*

- Led a multicultural team of 10+ people during peak harvest, supporting a 95% product acceptance rate through structured quality oversight.
- Managed irrigation systems on 50+ hectares of citrus.

- **Econexus Ventures Limited** — **Accra, Ghana**
  Production Team Lead — *Jan 2019-Oct 2020*
  - Led a cross-functional R&D team that took Ecogel, a bioethanol clean-cooking fuel, from formulation to market-ready production.

Selected Projects

- **GHG Inventory Tool (Scope 1 & 2), personal project**
  Live demo, GitHub
  - Python GHG Protocol inventory with per-gas calculations, dual Scope 2 reporting, Monte Carlo uncertainty, VSME/ESRS mapping, PDF and Excel outputs, and seven reduction options ranked by € per tonne.
- **Forest Carbon MRV Prototype, Kakum National Park, Ghana**
  GitHub
  - Pipeline structured to Verra REDD+ (VM0048), estimating about 12.5M tCO_2e (about 4.8M conservative, ±61.5%) from Sentinel-2, with Random Forest classification in Google Earth Engine (95.5% accuracy).
- **Green Roof Climate Data Pipeline and Dashboard, MSc thesis**
  GitHub
  - ETL pipeline that QC-filters and harmonizes multi-vendor sensor data into a 2.18M-record time-series warehouse (2020-2026), with a Streamlit dashboard for significance testing and energy-balance analysis.

Education

- **Technische Hochschule Bingen** — **Germany**
  MSc, Environmental Sustainability
  Thesis: "Green Roof Cooling Effectiveness Analysis".
- **University of Ghana** — **Accra, Ghana**
  BSc, Agricultural Engineering
- **AgroStudies Training Center** — **Netanya, Israel**
  Diploma in Agriculture (Horticulture)

Volunteering

- **Volunteer Data Analyst**, Net Zero Tracker (since 2023): documents the net zero commitments of 50+ companies and countries.
- **Environmental Volunteer**, Service Civil International (SCI Germany), Upper Middle Rhine Valley: landscape restoration and erosion control in vineyard terrain.

Certifications

- OPF Academy (Cohort 6): Net Zero Strategies, GHG Accounting, Sustainability Frameworks ▪ Google Data Analytics Professional Certificate ▪ BCG Climate & Sustainability Job Simulation, Forage ▪ Green Digital Skills, INCO Academy

Languages

- English (fluent); German (A2, actively learning).

Publications

- Panferov, O., Hietel, E., Rössner, U., Alef, J., & **Newton, J.** (2026). Rainwater-irrigated greenroofs as climate change adaptation in urban areas. EGU General Assembly 2026, BG10.12. doi:10.5194/egusphere-egu26-18880